CAPSTONE PROJECT

RideShift

Smart Pickup, Cheaper Rides

A Location-Based Fare Optimization App

Naiska Buyandalai

George Washington University

Advisor: Professor YuXiao Huang

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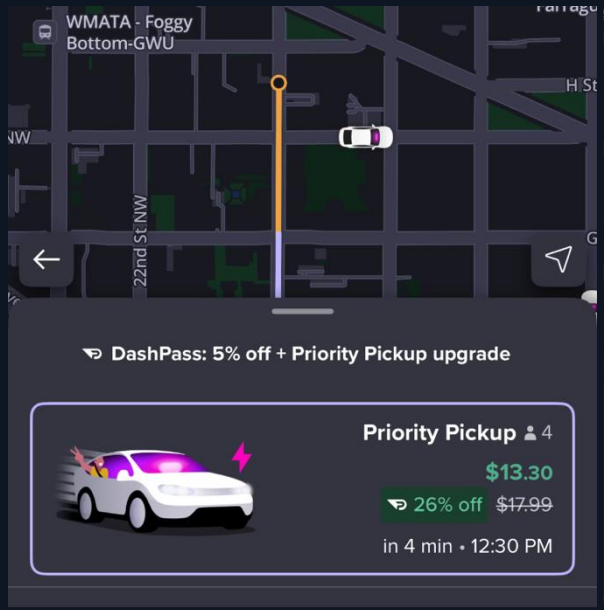
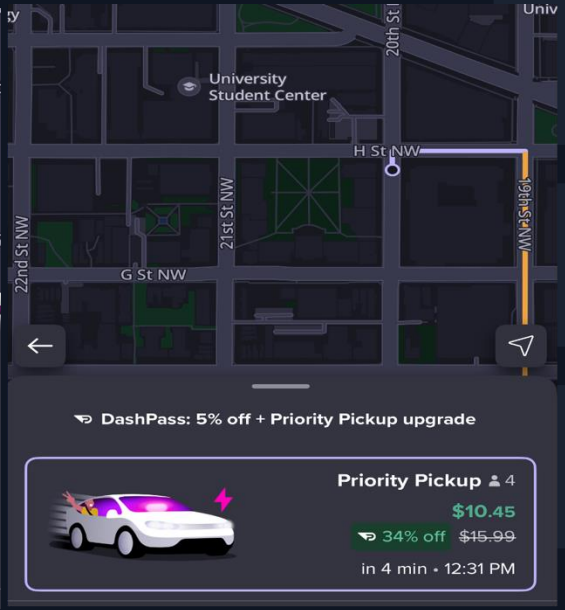
RideShift

A Location-Based Fare Optimization App

Smart Pickup, Cheaper Rides

THE PROBLEM





Up to 15-20%

fare variation within walking distance

Zero Tools

exist to help riders optimize pickup spot

~5 Min Walk

is all it takes to unlock cheaper fares

Solution & Tech Stack

HOW IT WORKS

1. **Enter Pickup & Destination**
   Address autocomplete via Apple MapKit
2. **Candidate Generation**
   Sample drivable road points within 5-min walk
3. **Beam Search Optimization**
   Search over candidates using fare estimation
4. **Results & Walking Route**
   Ranked savings & interactive map directions

TECH STACK

FRONTEND
SwiftUI & MapKit
iOS native app with interactive maps,
auto complete search engine, walking route overlays

BACKEND
Python
parallel processing, geocoding & isochrone caching

ALGORITHM
Beam Search & KD-Tree
Optimized candidate exploration with
fare API & DC formula fallback

APIS & DATA SOURCES

- **OpenRouteService**
  Geocoding + Isochrones
- **Overpass / OSM**
  Road Network Data
- **RapidAPI Taxi Fare Calculaor**
  Live Fare Estimates
- **DC DFHV Formula**
  Fare Fallback

Live Demo

WHAT YOU'LL SEE

Address Entry
Auto complete search engine for pickup & destination using Apple MapKit search

Optimization Run
Beam search evaluates nearby candidates in seconds

Fare Comparison
Side-by-side original vs. optimized fare with savings shown

Walking Route
Interactive map with walking directions to the better pickup spots

Ranked Results
Multiple alternatives sorted by fare with walk time tradeoffs

End-to-End Flow
Full user journey: search → optimize → walk → save

RideShift: Smart Pickup, Cheaper Rides

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